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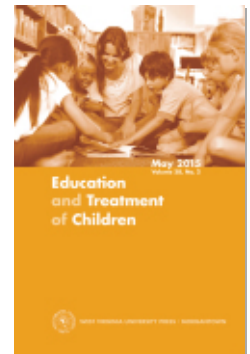
Teaching Socially Expressive Behaviors to Children with Autism Through Video Modeling

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Teaching Socially Expressive Behaviors to Children with Autism Through Video Modeling

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Abstract

Children with autism often lack complex socially expressive skills that would allow them to engage others more successfully. In the present study, video modeling was used to promote appropriate verbal comments, intonation, gestures, and facial expressions during social interactions of three children with autism. In baseline, the children rarely displayed any of the target behaviors. In treatment, each child watched a videotape of two persons interacting in a play setting. One person acted as the therapist and presented the social cues. The second acted as the child, and provided models of appropriate responses including verbal comments, intonation, gestures, and facial expressions. Results indicated that video modeling led to rapid acquisition of socially expressive behaviors. All three children reached criterion for all four target behaviors after viewing the video only three or four times. The children also displayed generalization of these socially expressive behaviors in probes across setting, stimuli, and persons.

KEYWORDS: affect, autism, facial expression, gesture, intonation, video modeling

Socially expressive behaviors enable children to share in a joined experience in both verbal and nonverbal ways. For example, when shown a new toy, a child might exclaim to his friend, "Wow, that toy is really neat!" In addition to a verbal response, this act involves more subtle forms of social behavior such as affect and gestures (Buffington, Krantz, McClannahan, & Poulson, 1998; Gena, Couloura, & Kymissis, 2005). The complexity of socially expressive behaviors makes them especially difficult for children with autism. First, most children with autism do not display interest in other people, and prefer to engage in

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solitary activities (Schreibman, 2005; Carter, Davis, Klin, & Volkmar, 2005). Second, many children with autism display inadequacies in most behaviors required for typical social expressions including verbal commenting, intonation, gestures, and facial expressions (Carter et al., 2005; Filipek et al., 1999; Rutter, 1978; Schreibman & Charlop-Christy, 1998).

Many interventions are aimed at teaching children with autism to speak. However, approximately 50% of all children with autism remain nonverbal (National Research Council, 2001; Schreibman, 2005). Furthermore, verbal children with autism often display perseverative, rote, and pedatonic speech (Filipek et al., 1999; Ricks & Wing, 1975). These children seldom use speech for social reasons, and tend to only use speech to make requests or to repeat their favorite phrases in inappropriate contexts (Schreibman, 2005). This is in stark contrast to neurotypical children, who use social referencing, speech, gestures, and other socially expressive behaviors to interact with and obtain information from their environment (Bornstein & Lamb, 1999). Thus, appropriate verbal commenting during social interactions is a suggested treatment target.

Intonation is also an important component of socially expressive behaviors. Although many children with autism can learn to talk, their speech is often idiosyncratic and monotone (Charlop & Haymes, 1994; Kanner, 1946). Researchers have found that when infants with autism babble, they lack inflection (Ricks & Wing, 1975). These deficits remain as children with autism age. High functioning persons with autism display impaired performance on prosody tasks (Peppe, McCann, Gibbon, O'Hare, & Rutherford, 2007), and are more likely to display residual articulation distortion errors, uncodable utterances, and inappropriate use of phrasing, stress, and resonance (Shriberg et al., 2001) than typical developing controls. Although intonation deficits are well documented, research does not indicate that these deficits can be successfully addressed by interventions.

As previously stated, socially expressive behaviors also include nonverbal components such as gestures and facial expressions. In neurotypical children, gestures develop early in life as a primary mode of communication that is associated with the later development of spoken language (Buffington et al., 1998). The development of gestures in children with autism is marked by both qualitative and quantitative impairments. Infants later diagnosed with autism use a more restricted range of gestures than their neurotypical peers (Colgan et al., 2006). Furthermore, during free play observations, children with autism use less expressive gestures than control children (Attwood, Frith, & Hermelin, 1988). Several studies have attempted to address

these impairments by teaching gestures to children with autism. Buffington and colleagues (1998) used modeling and prompting strategies to teach three children with autism to use gestures in combination with verbal speech in attention-directing, affective, and descriptive situations. More recently, Ingersoll, Lewis, and Kroman (2007) used reciprocal imitation training (RIT) to teach descriptive gestures to children with autism. In addition to learning to imitate gestures, the participants also spontaneously used the target gestures in the teaching environment.

Facial expressions are also important nonverbal communicative behaviors. When used appropriately, facial expressions display affective states, and are essential to social-communicative interactions (Gena, Krantz, McClannahan, & Poulson, 1996). Children with autism consistently show difficulty displaying appropriate affective behaviors (Koegel & Koegel, 1995; Schreibman, 1988). However, children with autism can learn to imitate facial expressions when the presentation of the stimuli is slow (Tardiff, Laine, & Rodriguez, 2007).

The present study combines previous research by teaching all four components of socially expressive behaviors (verbal comments, intonation, gestures, and facial expressions) to children with autism through the use of video modeling. In video modeling, children observe videotapes of the target behaviors and are then told to practice what they saw on the videos. Video modeling has been successfully used with children with autism to teach a variety of skills including reducing problem behaviors and increasing functional living skills, perspective taking, and social-communicative behaviors (see Delano, 2007 and McCoy & Hermansen, 2007 for reviews of video modeling with persons with autism). Of relevance to this study, many researchers have used video modeling to teach communication to children with autism. Examples include increasing requesting behaviors (Wert & Neisworth, 2003), spontaneous greeting behaviors (Charlop-Christy, Le, & Freeman, 2000), conversational speech (Charlop & Milstein, 1989; Charlop-Christy et al., 2000; Sherer et al., 2001), social initiations (Nikopoulos & Keenan, 2007), and play-related comments directed towards siblings (Taylor, Levin, & Jasper, 1999). Although most of these studies only targeted the children's verbal speech, one video modeling study targeted two communicative behaviors: verbal speech and facial expressions (Gena et al., 2005). Gena and colleagues taught affective behaviors to three preschoolers with autism in a study that compared video modeling with in-vivo modeling. Both of the methods were effective in teaching affective behaviors in sympathy, appreciation, and disapproval situations.

The present study extends previous research by including

measurements of intonation and gestures as well as verbal comments and facial expressions. It was hypothesized that video modeling would increase the four target behaviors (verbal comments, intonation, gestures, and facial expressions) in all three participants, and that these socially expressive behaviors would generalize across persons, setting, and stimuli.

Method

Participants

Three boys with autism who attended an after-school behavior therapy program participated in this study. All of the children were diagnosed with Autism Spectrum Disorder by two independent agencies according to the criteria of the DSM-IV-TR (American Psychiatric Association, 2000). Diagnostic measures included the Autism Behavior Checklist (ABC; Krug, Arick, & Almond, 1980), Childhood Autism Rating Scale (CARS; Schopler, Reichler, & Renner, 1988) and the Gilliam Autism Rating Scale (GARS; Gilliam, 1995). The children were between the ages of 7 and 11 and had attended the behavior therapy program from 3 months to 5 years. All three participants were chosen for this study because they displayed deficits in socially expressive behaviors (e.g., appropriate verbal comments, intonation, gestures, facial expressions) as reported and observed by the children's therapists, teachers, and parents.

At the onset of the study, Eddie was 11 years, 9 months old, and had an overall Vineland Adaptive Behavior Composite (Sparrow, Ball, & Cicchetti, 1984) of 71. He exhibited good verbal skills, but did not always use his verbalizations in an appropriate social context. For example, he would often ask questions about perseverative topics (e.g., "What is the color of your carpet?"). He rarely used gestures such as pointing or nodding. He made facial expressions (e.g., grimacing) while talking, but did not display contextually appropriate facial expressions (e.g., he did not smile when he was happy, and his face was emotionless when his stomach hurt). His intonation was poor, and he spoke in brief phrases with a monotone or grunting voice.

Mark was 8 years, 5 months old, and had a mental age of 6 years, 8 months, as measured by the Peabody Picture Vocabulary Test (Dunn & Dunn, 1981). Mark had strong verbal skills, but lacked the use of gestures, facial expressions, and intonation. He could answer simple questions and could ask questions about perseverative topics (e.g., "Do you know that video game?"), but rarely commented about activities or items around him. He displayed flat affect, and similarly spoke with a flat tone of voice even when reporting various emotions (e.g., happy or sad).

Terry was 7 years, 1 month old, and had a mental age-equivalent score of 4 years, 4 months, as measured by the Vineland Social Maturity Scale (Sparrow, Ball, & Cicchetti, 1984). He would imitate phrases with prompting, but only used spontaneous speech to request desired items. He rarely made verbal comments and was extremely soft-spoken with a "sing-song" tone of voice. Terry rarely used gestures including pointing. Although he used some facial expressions, such as adopting a grim look with furrowed eyebrows when he was upset, he rarely used a combination of appropriate gestures, facial expressions, and speech.

Materials

A video was made for each child. Each video consisted of three scenarios that were repeated three times in an alternating order, for a total of nine scenarios. Each scenario presented a discriminative stimulus representing a situation in which it was appropriate for the child to make a verbal comment, along with an intonation, gesture, and facial expression. For example, in Terry's video, one of the discriminative stimuli was being shown a large toy. In this situation, the target responses included verbalizing "That's a **BIG** (ball)!" with an exclamation, gesturing with his arms wide, and raising his eyebrows. Table 1 outlines the discriminative stimuli and appropriate responses for each child.

Two familiar adults were featured in each of the videotapes. For each scenario, one adult presented the discriminative stimulus and a second adult, acting as the child's model, then exhibited the appropriate response using the target verbal comment, intonation, gesture, and facial expression. After the adult depicting the child model exhibited the target responses, the second adult would provide the naturally occurring responses. For example, if the child said "Stop tickling", the adult would stop tickling. The models always faced the camera when delivering their lines and displayed the target behaviors at an exaggeratedly slow pace as suggested by previous research (e.g., Charlop & Milstein, 1989; Charlop, Gilmore, & Chang, 2008). Videos were shot using third person scene perspective where the learner is depicted acting in the video and each video lasted approximately 90 seconds.

Since video modeling is an individualized treatment, it was assumed that individual videos would be necessary. For example, Terry's highly preferred toy for the "that's cool" segment was different than the toy used in Mark's "that's cool" segment given their different preferences. When possible, the same video segment was used for each child being taught the same target behavior. For example, the same "stop tickling" segment was used on Mark and Eddie's videos.

Table 1
Discriminative Stimuli and Target Responses for Each Child

Child	Discriminative stimulus		
Eddie	Tickling or deny access to preferred toy	Child shown a preferred toy	Child makes a basket or knocks down bowling pins
Verbalization	Stop it!	That's cool!	I got it!
Intonation	Loud/Authoritative	Exclamation	Exclamation
Gesture	Raise arms wide	Hands up/Palms out	Thumbs up
Facial expression	Squint eyebrows	Raise eyebrows	Smile
Child	Discriminative stimulus		
Mark	Child shown a large toy	Tickling or deny access to preferred toy	Drop an object
Verbalization	That's a big (ball)!	Stop it!	I'll get it!
Intonation	Exclamation	Loud/Authoritative	Exclamation
Gesture	Raise arms wide	Hands up/Palms out	Thumbs up
Facial expression	Raise eyebrow	Squint eyebrows	Smile
Child	Discriminative stimulus		
Terry	Child makes a basket or knocks down bowling pins	Child shown a large toy	Child shown a preferred toy
Verbalization	I got it!	That's a big (ball)!	That's cool!
Intonation	Exclamation	Exclamation	Exclamation
Gesture	Raise arms	Raise arms wide	Point
Facial expression	Smile	Raise eyebrows	Raise eyebrows

Setting

The children watched the videos in a work room (2.9 x by 2.9 m) at their after-school program. The room contained a small table, a VHS recorder, a 19-in. color television monitor, and a chair for the child to sit directly in front of the television. Baseline and treatment play sessions took place in a playroom (2.6 x 4.4 m). The playroom contained a table, two child-sized chairs, and the toys used in the training sessions. Peer probes were held in the same free-play room, and adult generalization probes were held in a nearby room that was unfamiliar to the child.

Design

A multiple baseline design across children was used to assess the effects of the video modeling treatment on the four target skills. After baseline, video modeling was then presented to each child. Treatment play sessions were held to assess acquisition of the target behaviors. Adult generalization probes were taken during baseline and post-treatment. Peer probes were taken post-treatment as additional information. As baseline peer probes were only available for Mark, information from these probes do not allow for conclusions to be made regarding generalization to peers.

Procedure

Baseline. The children's verbal comments, intonations, gestures, and facial expressions were assessed in response to the discriminative stimuli. Eddie had 5 baseline sessions, Mark had 7 baseline sessions, and Terry had 10 baseline sessions. At the beginning of each session, the experimenter initiated the interaction by presenting one of the discriminative stimuli, representing a situation in which it was appropriate for the child to respond with a social expression. The child was then given 5-seconds to emit the four appropriate responses (target verbal comment, intonation, gesture, and facial expression). If no response occurred within 5-seconds, the trial was terminated and the child and experimenter returned to their previous play activities. If the participant gave an appropriate response (e.g., "I GOT it!" after making a basket), the response was reinforced with an appropriate reciprocal response (e.g., "WAY to GO!"). Each 10-minute play session contained 9 opportunities for the child to display the appropriate target behaviors (3 times for each of the 3 stimuli).

Video Modeling. At the beginning of each session, the child was asked to sit quietly and watch the television. The experimenter sat next to the child to ensure that the child attended to the video. If the child looked away from the video, the experimenter would say, "Pay

attention." After watching the video two consecutive times, the child was immediately assessed in a play session.

The experimenter and child walked to the playroom, where the experimenter initiated interaction by presenting a toy or activity, as shown in the video and conducted in baseline. The child was given 5-seconds to emit the appropriate response. As in baseline, if no response occurred within 5-seconds, the trial was terminated and the child and experimenter returned to their previous play activities. If the participant gave the target response or a similar appropriate response (e.g., "How cool!" after being shown a new object), the response was reinforced with an appropriate reciprocal response (e.g., child was given object to play with). The play session then proceeded as in baseline (i.e., free play with intermittent opportunities to display socially expressive behaviors) for a total of 9 trials each session.

The mastery criterion was met when a child displayed the appropriate four responses (i.e., verbal comment, gesture, facial expression, and intonation) in 7 out of the 9 trials for two consecutive play sessions. If the child did not meet criterion after the first two sessions, he was shown the video again. This procedure of showing the video and then assessing behaviors occurred until the child met criterion for all target responses (as in Charlop & Milstein, 1989).

Generalization probes (across person, setting, and stimuli). During baseline and post-treatment, generalization probes were taken with an unfamiliar person in an unfamiliar room set up for play. The stimuli used in these sessions were similar, but not identical to those used in the playroom baseline and treatment sessions. The procedure followed that of the playroom baseline and treatment sessions. During baseline, Eddie, Mark, and Terry had two, three, and four generalization play sessions, respectively. In addition, each child had two post-treatment generalization play sessions.

Peer generalization probes. In order to further assess generality of the participants' responses following treatment, two probes were taken for each child with a peer. Mark acted as Eddie's peer. Stan, an 8-year old friend, acted as Terry's peer, and Lisa, Mark's 5-year old sister, acted as Mark's peer. During these sessions, the experimenter instructed the peer to present each discriminative stimulus, for a total of nine trials. For Mark, peer probes were also taken during baseline, following the same procedure.

Social Validity of Treatment Effects

Thirty-seven undergraduate college students who had no formal experience with children with autism rated the children's behavior after watching an edited videotape. The videotape consisted of six

randomly selected segments with three of the segments from each child's baseline and three of the segments from each child's post-treatment sessions. Each segment consisted of three interactions between the child and the experimenter. After each segment, the videotape was paused, and the undergraduates filled out a questionnaire about what they saw. This procedure was repeated until the undergraduates rated all six segments. The questionnaire included ten items that assessed the child's use of verbal comments, intonation, gestures, and affect as well as how "likeable" the child seemed, using a 5-point Likert-type scale (see Table 2). The raters were not informed whether they were observing segments from baseline or post-treatment.

Table 2
Social Validity Results

Item	Child		
	Eddie	Mark	Terry
1. The child talks appropriately about what he sees.	$p < .001$	$p < .01$	$p < .001$
2. The child speaks clearly.	$p < .001$	n.s.	$p < .001$
3. The child uses his hands to express himself.	$p < .001$	$p < .001$	$p < .001$
4. The child has hand movements that add to his ability to communicate	$p < .001$	$p < .001$	$p < .001$
5. The child's voice expresses emotions (e.g., excitement, anger, etc).	$p < .001$	$p < .001$	$p < .001$
6. The child's voice volume/expression adds to his ability to communicate.	$p < .001$	$p < .01$	$p < .001$
7. The child's facial expressions are appropriate to the situation.	$p < .001$	n.s.	$p < .001$
8. The child's facial expressions corresponds with the child's speech.	$p < .001$	n.s.	$p < .001$
9. I would baby-sit this child.	$p < .05$	n.s.	n.s.
10. The child is likeable.	$p < .001$	n.s.	$p < .001$

Scoring

Videotapes of each session were coded by the experimenter. For each session, the children received a separate score for appropriate verbal comments, intonations, facial expressions, and gestures by dividing the number of correct responses by 9, the total number of trials. A trial was coded as correct if the child displayed the target response, or a similar appropriate response, within 5 seconds of the discriminative stimulus (see Table 3 for an outline of the operational definitions). For example, if the target verbal comment was "I made it!", "I got it!" would also be scored as a correct verbal comment.

Interobserver agreement

Following the same procedure for scoring, a second rater independently scored one-third of all of the sessions. This rater was trained by reviewing operational definitions of the target behaviors and viewing the first baseline session for each participant. Agreement was calculated by dividing the total number of agreements by the total number of disagreements, and then multiplying this number by 100. Interobserver reliability averaged across all children was 90% for all four-response categories throughout all conditions of the study.

Procedural Reliability

In order to further control for potential bias in this study, procedural reliability was calculated for one-third of all baseline, treatment, and generalization sessions for each child. An outside observer watched these videos, selected at random, and assessed if the stimuli were presented, if there were any extraneous prompts, and if each trial was presented in a consistent manner. Procedural reliability was maintained at a level above 90% for all sessions.

Results

The percentage of opportunities correct during each session is shown for each target response: verbal comments (Figure 1), intonations (Figure 2), gestures (Figure 3), and facial expressions (Figure 4). Very few target responses were made during baseline. After video modeling was introduced, all three children met criterion for all four target behaviors. Because each child met criterion for some behaviors faster than others, for several of the target behaviors data continued to be taken in the treatment sessions after criterion was met. For example, Mark met criterion for verbal comments quicker than he met criterion for facial expressions. Therefore, although criterion was met for verbal comments after two sessions, data was still taken on verbal comments in sessions three and four. The children's display of socially

Table 3
Operation Definitions of Dependent Measures

Scenario	Target behavior	Operation definition
Tickling/Deny access	Verbalization: "Stop it"	Says "stop it" or similar content (i.e., "stop", "stop doing that", "don't do that")
	Intonation	Loud/Authoritative, negative valence, word "stop" or similar meaning word spoken louder than others
	Gesture	One or two hands up, palm vertical and facing adult, elbow straight or bent accepted
	Facial expression	Eyebrows moved closer together than in resting position (i.e., furrowed)
Shown a preferred toy	Verbalization: "That's cool!"	Says "that's cool" or similar content (i.e., "that is cool", "very cool", "cool")
	Intonation	Exclamation, positive valence, word "cool" or similar meaning word spoken louder than others
	Gesture	Point with index finger or full hand extended towards object, or reaching for object with arm extended
	Facial expression	Eyebrows raised at least ¼ inch from resting position
Makes a basket/ knocks down pins	Verbalization: "I got it!"	Says "I got it" or similar content (i.e., "I win", "I made it", "I did it")
	Intonation	Exclamation, positive valence, word "got" or similar meaning word spoken louder than others
	Gesture	Arms raised above head, either straight arms or elbows bent
	Facial expression	Smile, lips turned up on side at least 1/8 inch from resting position, teeth visible
Shown a large toy	Verbalization: "That's a big (ball)!"	Says "That's a big (ball)" or similar content (i.e., "That's a big one", "That's so big", "Big one")
	Intonation	Exclamation, positive valence, word "big" or similar meaning word spoken louder than others
	Gesture	Arms raised or extended in front of body, with at least 1 foot in between hands
	Facial expression	Eyebrows raised at least ¼ inch from resting position

(continued overleaf)

Table 3 (continued from p. 381)
Operation Definitions of Dependent Measures

Scenario	Target Behavior	Operation definition
	Verbalization:	Says "I'll get it" or similar content (i.e., "I got it", "I'll do it", "I'll get that")
	Intonation	Exclamation, positive valence, word "I'll" or similar meaning word spoken louder than others
	Gesture	Thumbs up with thumb raised and other fingers tucked under, using one or both hands
	Facial expression	Smile, lips turned up on side at least 1/8 inch from resting position, teeth visible

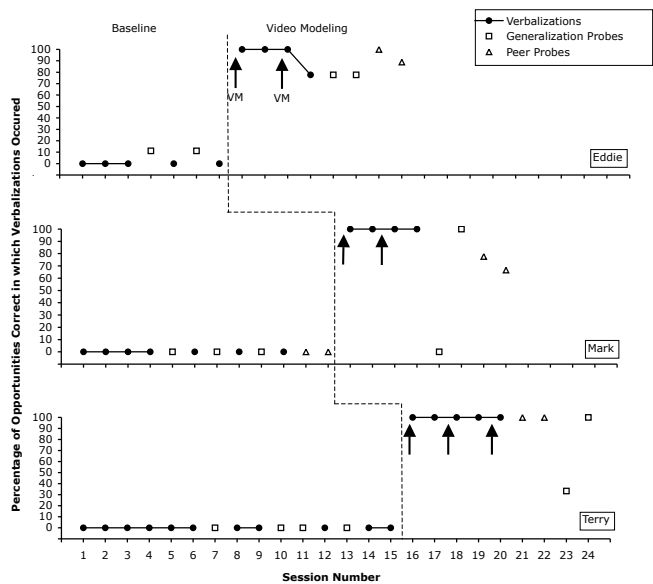


Figure 1. Percentage of opportunities in which appropriate verbal comments occurred during baseline, treatment, and generalization play sessions for all three children.

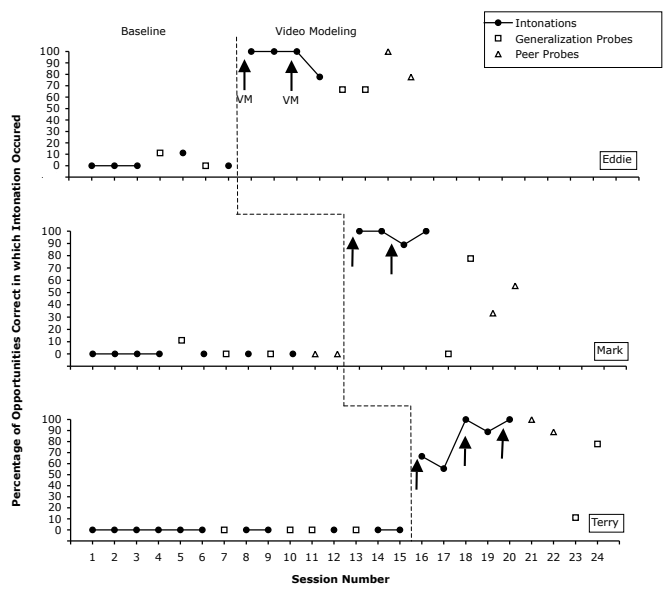


Figure 2. Percentage of opportunities in which appropriate intonations occurred during baseline, treatment, and generalization play sessions for all three children.

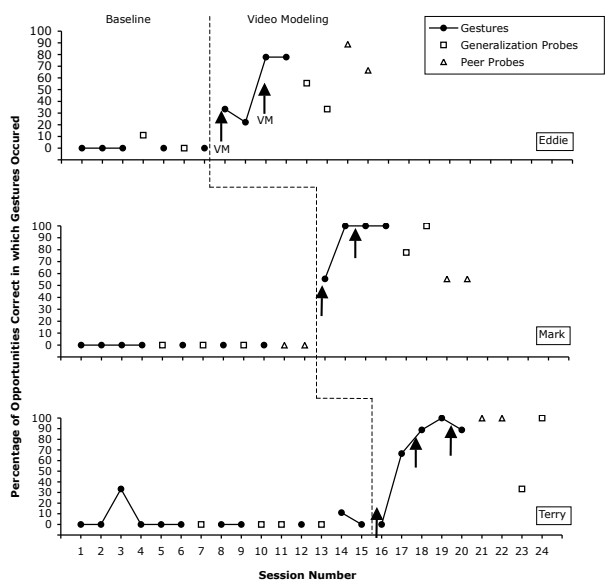


Figure 3. Percentage of opportunities in which appropriate gestures occurred during baseline, treatment, and generalization play sessions for all three children.

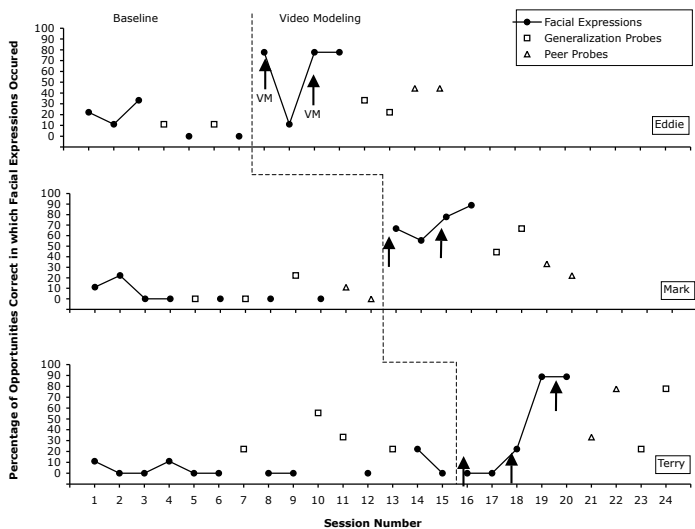


Figure 4. Percentage of opportunities in which appropriate facial expressions occurred during baseline, treatment, and generalization play sessions for all three children.

expressive behaviors during post-treatment generalization probes, and peer probes varied with child and target response. However, all children displayed increases in social behaviors when comparing post-treatment generalization percentages with baseline percentages.

Verbal Comments

During baseline sessions, none of the children consistently displayed the target verbal comments. After video modeling was implemented, all of the children displayed rapid increases in appropriate verbal comments, reaching criterion after only two sessions (see Figure 1). Eddie (top panel) did not display any appropriate verbal comments during baseline sessions with one appropriate comment during the baseline adult generalization probes. After treatment, he immediately met the criterion in the first two play sessions. During his post-treatment generalization sessions, Eddie consistently displayed the target verbal comments in 7 out of 9 opportunities with similar findings during the peer probes (verbal comments in 9 out of 9 and 8 out of 9 opportunities). Mark (middle panel) did not display any

appropriate verbal comments in baseline play sessions, baseline adult probes, or baseline peer probes. Immediately following treatment, Mark reached criterion over two consecutive sessions, displaying the target behavior in 9 out of 9 opportunities. Mark did not display any of the target verbal comments in the first post-treatment generalization session, but displayed an appropriate response on every trial in the second session. Lastly, in the post-treatment peer sessions with his sister, Mark made an appropriate comment in 7 out of 9 and 6 out of 9 opportunities. Terry (bottom panel) did not display any appropriate verbal comments in baseline play sessions or adult probes. Terry immediately met the criterion after treatment and performed perfectly in the peer probes (9 out of 9 verbal comments). In the post-treatment generalization probes, Terry made appropriate verbal comments in 3 out of 9 and 9 out of 9 opportunities.

Intonation

None of the children consistently displayed appropriate intonation during baseline. After video modeling was introduced, the children quickly reached criterion, requiring from two to four sessions (see Figure 2). Eddie (top panel) displayed low levels of appropriate intonation during baseline sessions and generalization probes. Eddie immediately met the criterion after treatment was introduced (9 of 9 opportunities in the first two treatment sessions). He used appropriate intonation in 6 out of 9 opportunities during both post-treatment generalization sessions, and in 9 out of 9 and 7 out of 9 opportunities during the post-treatment peer sessions. Mark (middle panel) had no appropriate intonations during baseline sessions or probes with his sister and one appropriate intonation during a baseline generalization probe. After treatment, Mark immediately met the criterion with appropriate intonation in 9 out of 9 opportunities in the first two treatment sessions. In the first post-treatment generalization probe, Mark did not display any appropriate intonation; however, in the second post-treatment generalization probe he displayed appropriate intonation in 7 out of 9 opportunities. In the peer generalization probes with his sister, Mark displayed appropriate intonation in 3 out of 9 and 5 out of 9 opportunities. Although responding was not maintained at amounts equivalent to those during treatment, Mark still demonstrated appropriate intonation during the post-treatment generalization probes at higher levels than during baseline generalization probes.

Terry (bottom panel) did not display any appropriate intonation during baseline sessions or baseline generalization probes. He reached the criterion after four treatment sessions, demonstrating appropriate intonations on all 9 opportunities in the third treatment session and

in 8 out of 9 opportunities in the fourth treatment sessions. During the two post-treatment peer generalization probes, Terry demonstrated appropriate intonation in 9 out of 9 and 8 out of 9 opportunities. Lastly, in the first post-treatment generalization probe, Terry used the correct intonation in 1 out of 9 opportunities, and in 7 out of 9 opportunities in the second post-treatment generalization probe.

Gestures

None of the children consistently displayed appropriate gestures during baseline(see Figure 3).. After video modeling was introduced, the children met criterion after three to four treatment sessions. In addition, all of the children displayed higher levels of appropriate gestures during post-treatment generalization probes than during baseline generalization probes. Eddie (top panel) did not use any appropriate gestures during baseline sessions and used only one appropriate gesture during one baseline generalization probe. After treatment was introduced, Eddie met the criterion for gestures in four treatment sessions. Eddie displayed appropriate gestures in 5 out of 9 and 3 out of 9 opportunities in the post-treatment generalization probes, and in 8 out of 9 and 6 out of 9 opportunities in the post-treatment peer probes. Mark (middle panel) did not use any appropriate gestures during baseline sessions, baseline generalization probes, or baseline probes with his sister. Mark met the criterion for gestures after three treatment sessions. He displayed appropriate gestures in 7 out of 9 and 9 out of 9 opportunities in the two post-treatment generalization probes, and in 5 out of 9 opportunities for both of the generalization peer probes. Terry (bottom panel) used a few appropriate gestures throughout baseline but did not display any appropriate gestures during baseline generalization probes. Terry reached the criterion for appropriate gestures after four treatment sessions and did very well in post-treatment peer probes (9 of 9) and in one of the two post-treatment generalization probes (i.e., 3 of 9, 9 of 9).

Facial Expressions

Interestingly, facial expressions were the most frequently emitted correct behavior during baseline (see Figure 4), although no child reached criterion in baseline. All of the children reached criterion after video modeling was introduced, with a range of four to five treatment sessions necessary to reach criterion. Eddie (top panel) displayed low levels of appropriate facial expressions during baseline sessions and during baseline generalization probes. He reached criterion for the target behavior after four treatment sessions with low to moderate performance in the two post-treatment generalization probes and the

post-treatment peer probes. Mark (middle panel) also displayed low levels of appropriate facial expressions during baseline sessions, baseline generalization probes, and baseline peer probes with his sister. Mark reached criterion for facial expressions after four treatment sessions. Similarly to Eddie, Mark's levels of appropriate facial expressions during post-treatment generalization probes and peer probes were lower than anticipated but still increased somewhat over baseline. Terry (bottom panel) also displayed some appropriate facial expressions during baseline sessions and during baseline adult generalization probes. Terry reached criterion for facial expressions after five treatment sessions. Terry displayed appropriate facial expressions in 3 out of 9 and 7 out of 9 opportunities in the two post-treatment generalization probes, and in 2 out of 9 and 7 out of 9 opportunities in the two post-treatment peer probes.

Social Validation

Paired sample *t*-tests were conducted to examine differences in naïve observers' ratings of baseline and post-treatment segments for each question in the Social Validity Questionnaire (see Table 2). In general, observers rated the children's socially expressive behaviors as more appropriate during post-treatment segments than baseline segments. For Mark, the observers' scores differed on 5 out of 10 of the items. For Eddie, the observers' scores differed on all 10 items, and for Terry the observers' scores differed on 9 out of 10 of the items.

Discussion

The results of the present study indicate that video modeling successfully promoted socially expressive behaviors in three children with autism. All of the children in this study reached criterion for acquisition (7 out of 9 opportunities, or approximately 80% for two consecutive sessions) for all four target responses: verbal comments, intonation, gestures, and facial expressions. All three children displayed rapid acquisition of the target behaviors. Eddie and Mark reached criterion for all four target responses after four treatment sessions, and watching the video three times. Terry reached criterion for all four target responses after five treatment sessions, and watching the video four times. These acquisition rates are similar to those in previous studies that also used video modeling (Charlop & Milstein, 1989; Charlop-Christy et al., 2000). This study may be viewed as a relatively "pure" intervention in that no other treatment components such as rehearsal, prompting, or additional reinforcers were used. The only reinforcers that were in effect were the natural social reinforcers that occurred as a function of the child's behaviors.

There are several reasons why video modeling may lead to rapid acquisition in children with autism. First, video modeling incorporates a reinforcing activity into teaching. Children often enjoy watching television and videos. If watching a video is a reinforcing activity, then children with autism may be more motivated to attend to the video and more likely to learn or imitate the modeled activity (Charlop-Christy et al., 2000). Second, video modeling takes advantage of the visual strengths of children with autism (McCoy & Hermansen, 2007). It has been well documented that children with autism may have enhanced abilities processing visual stimuli (Bryan & Gast, 2000; Hermelin, 1976), and a variety of interventions that incorporate visual cues (e.g., PECS, picture activity schedules, and video modeling) have been found to be especially effective for this population (Charlop et al., 2008). Lastly, effective videos will focus the children's attention on the relevant cues (McCoy & Hermansen, 2007), making up for problems associated with stimulus overselectivity, or the failure to respond to multiple cues presented simultaneously. Children with autism often have difficulty attending to more than one stimulus at a time (Lovaas, Schreibman, Koegel, & Rehm, 1971), and if they attend to the "wrong" stimulus, acquisition of the target behavior may be delayed. However, by highlighting the relevant cues, video modeling helps children with autism focus on the important information (Charlop-Christy & Daneshvar, 2003).

This study also demonstrates the utility of video modeling when targeting multiple behaviors simultaneously. Gena and colleagues (2005) first used video modeling to teach children with autism to pair appropriate facial expressions with different verbalizations. The present study extends this work by demonstrating that one video can be used to teach four behaviors (verbal comments, intonation, gestures, and facial expressions). Not only is video modeling a relatively inexpensive intervention, but it may also be an efficient intervention when teaching complex behaviors. Instead of teaching each response category separately, video modeling allowed the children to learn all four responses simultaneously. By the end of the study, the children were engaging in complex behaviors that resembled the socially expressive behaviors of typically developing children.

Indeed, the social validation data provides further evidence that the children were demonstrating appropriate socially expressive behaviors after treatment. The naïve raters indicated that almost all of the children's verbal comments, intonations, gestures, and facial expressions were more appropriate after treatment than during baseline. Furthermore, the raters gave Eddie and Terry a higher likeable score after treatment than during baseline. This data confirms the idea that

socially expressive behaviors affect our perceptions of other people, and, therefore, are an important target for children with autism.

One of the strengths of this study is that it both trained for and measured generalization, an important consideration in treatments for children with autism (Schreibman, 1988; Stokes & Baer, 1977). The videos incorporated multiple exemplars for several of the stimuli (e.g., saying "I got it!" after making a basket and knocking down bowling pins). Also, all of the toys in the videos were from the children's natural environment (Charlop-Christy et al., 2000; Stokes & Baer, 1977). Although the generalization results varied for each child and behavior, at the end of treatment, all three children demonstrated all four behaviors across people, setting, and stimuli at higher rates than during baseline. One limitation of the present study is the lack of baseline peer probes for two of the three children. Therefore, interpretation of Eddie and Terry's post-treatment peer probes should be made with caution because there are no baseline probes for comparison.

In addition to generalization across people, setting, and stimuli, the children also displayed response variation. Terry generalized one of the appropriate verbal comments from "I got it" to "I made it!" after making a basket during his adult and peer generalization sessions. Mark exclaimed, "That's a really big box!" instead of the modeled comment ("That's a big box!"). Finally, Eddie demonstrated a novel gesture when he would make a basket, lifting up one leg in addition to raising both of his arms above his head, making him appear even more excited.

Although the children demonstrated generalization for each of their three specific discriminative stimuli, the present study does not indicate whether or not video modeling also increased the children's socially expressive behaviors in other situations. Future research should investigate the use of video modeling when targeting a range of socially expressive behaviors, and include generalization probes across untrained situations.

In conclusion, the present study adds to the literature by demonstrating that socially expressive behaviors can be taught to children with autism using video modeling. Further, multiple target behaviors were rapidly promoted using a single video. The investigation of training techniques to promote affect in children with autism has been largely under-represented in the field. Future research should be conducted to provide a better understanding of how to improve socially expressive behaviors in children with autism. In addition, it would also be interesting to determine the effects of a video modeling treatment with lower functioning children with autism and its effectiveness at teaching nonverbal communication skills to these children.

Note

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